```
  time ←
0
      #2
  in_time ← null (" v ∈
graph)                                                    #3
  out_time ← null (" v ∈ graph)
  for v in graph.vertices
do                                                       #4
    if in_time[v] == null
then                                                     #5
      (time, in_time, out_time) ← dfs(graph, v, time, in_time,
out_time)      #6
  return (in_time,
out_time)                                                  #7
```

#1 Method `dfs` takes a graph, traverses all of its vertices, and returns two dictionaries with the times at which each vertex was entered and exited.
#2 Initialize the “time” counter. This is needed to keep track of the order of discovery of vertices.
#3 Initialize the enter and exit times for each vertex to `null`.
#4 Iterate over all the vertices in the graph.
#5 If a vertex v’s in-time is still `null`, then it hasn’t been discovered yet.
#6 If so, start a dfs traversal starting from v.
#7 Once all vertices have been visited, return the enter and exit times.